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Supervision meets Self-Supervision: A Deep Multitask
Network for Colorectal Cancer Histopathological Analysis

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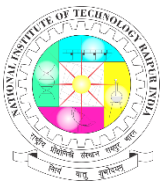


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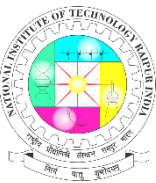
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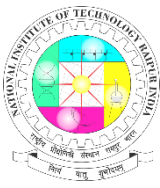


Background

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- The necessity of higher efficiency in colorectal cancer histopathological (CRCH) analysis from the tissue images along with the extraction of underlying features from those images has paved the way for deep learning based methods in the literature.
- Medical imaging-based histopathological analysis has been approached classically using traditional machine learning and handcrafted feature extraction, which, however, fail to capture complex underlying patterns within the image data.
- Deep learning methods, particularly CNNs, have gained popularity due to their capability in recognising salient and translationally invariant image features for robust classification and investigation.



Literature Survey

Colorectal Cancer Histopathology:

- Kather et al. [1] proposed the first multi-class texture analysis in colorectal tissue classification.
- Wang et al. [2] proposed deep learning based histopathological image classification with bilinear convolutional neural networks.
- Ciompi et al. [3] highlighted the importance of stain normalization in colorectal tissue classification with convolutional networks.
- Sabol et al. [4] proposed a method which had enhanced classifier performance using a fine-tuned CNN model for multi-class classification task.
- Other deep learning based works include transfer learning by Ohata et al. [5] and ensemble learning strategy by Ghosh et al. [6].



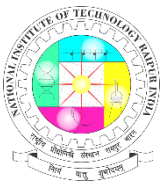
Literature Survey

Deep Metric Learning:

- Based on (dis)similarity learning between samples and has been used for representation learning in several domains such as offline signature verification (Bromley et al. [7]), medical imaging (Annarumma et al. [8]) and face recognition (Liu et al. [9]; Schroff et al. [10]), to name a few.

Self-Supervised Learning:

- SSL has gained tremendous popularity due to its capability to learn robust representations from unlabelled data. Primary directions of study include contrastive learning (Chen et al. [11]), generative/reconstruction based models (Atito et al. [12]; Deepak et al. [13]) and clustering approaches (Alwassel et al. [14]).



Problem Description

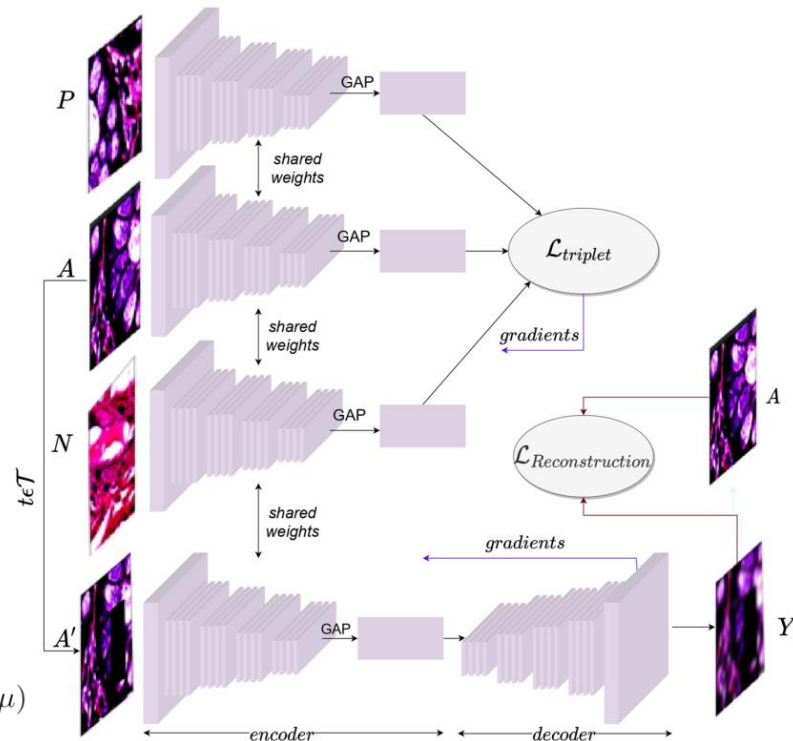
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- As of 2020, CRC is one of the leading causes of cancer related deaths in the world, accounting for around 11% of the cancer patients worldwide.
- The early detection of CRC can increase the survival rates by around 90% which explains the need for its early detection.
- An early detection and diagnosis is of paramount importance, which is however, limited due to insufficient medical practitioners available for large-scale histopathological screening.
- This demands for a reliable computer-aided framework that can automatically analyse histopathological slide images and assist pathologists in quick decision-making.
- To this end, we propose a novel deep learning framework that combines supervised learning with self-supervision for robust learning of histopathological features from colorectal tissue images.

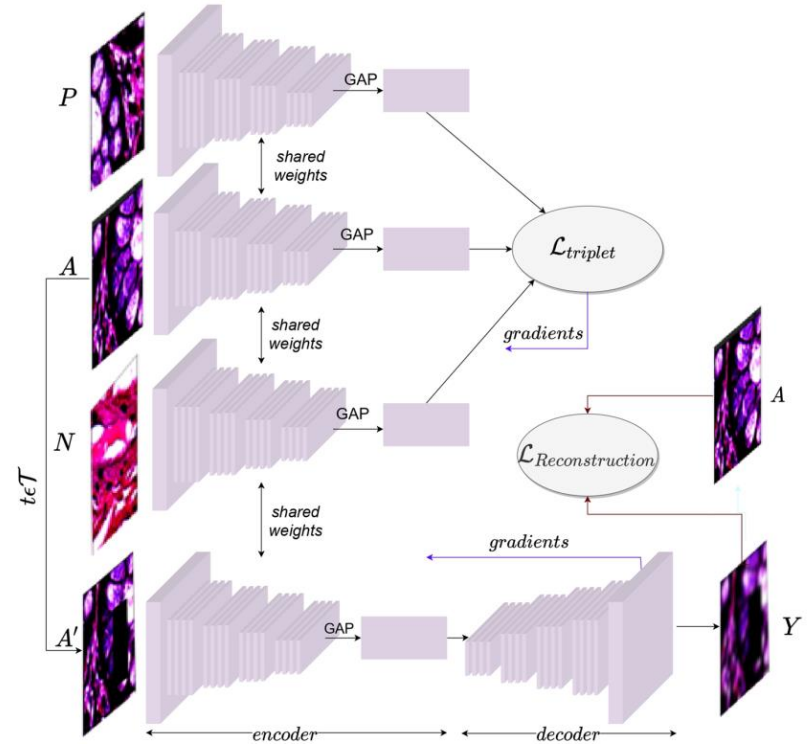
- We intend to combine supervised learning with self-supervision as a pre-training paradigm for robust CRCH image analysis.
- We use the metric learning framework to learn maximisation of intra-class similarities while distinguishing samples from a different class.
- Specifically, metric learning is used to minimise the distance metric between intra-class or “positive” pairs and maximise the distance between inter-class or “negative” pairs.
- The primary purpose is to learn discriminative embedding space, where samples from same class shall be closer while the ones from different classes shall be farther apart.

$$\mathcal{L}_{triplet} = \sum_{i=1}^K \max(0, \|\mathcal{F}(A^{(i)}) - \mathcal{F}(P^{(i)})\|^2 - \|\mathcal{F}(A^{(i)}) - \mathcal{F}(N^{(i)})\|^2 + \mu)$$



- Further, we have introduced a self-supervised image reconstruction module to enhance the learning of image-level texture features.
- For reconstruction, we have used the state-of-the-art U-Net decoder module (Ronneberger et al. [15]) excluding the skip connections, starting from an embedding vector into a series of unsampling layers which yield the output image dimensions.
- We form a corrupted version of the ‘anchor’ image by random spatial transformations such as blurring or dropping pixels, which is then used to generate the reconstruction.
- We have used mean squared error(MSE) as the reconstruction loss, employed between the output and original anchor image to train this branch.

$$\mathcal{L}_{reconstruction} = \frac{1}{M} \sum_{i=1}^K \sum_{j=1}^M \|A_j^{(i)} - Y_j^{(i)}\|_2^2$$

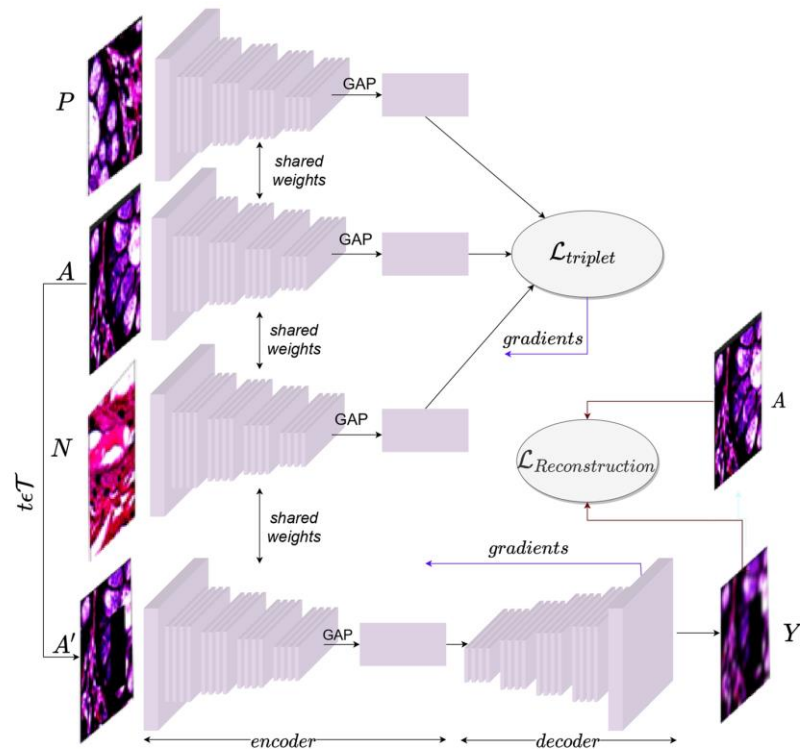


Combining the two losses, we define our training objective as:

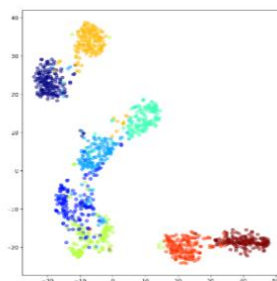
$$\mathcal{L}_{train} = \lambda \cdot \mathcal{L}_{triplet} + \mathcal{L}_{reconstruction}$$

λ is a hyperparameter to balance the losses.

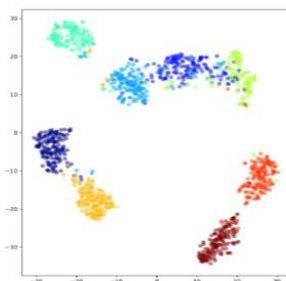
- Once the network is trained, the frozen encoder is used to extract image features, which are used to train a SVM classifier for the final classification.



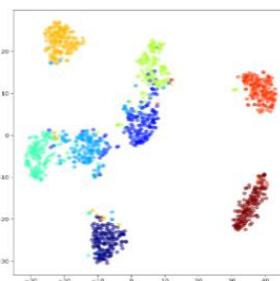
- The following figures illustrates the t-distributed stochastic neighbourhood embedding (t-SNE) plots at different intervals during the training process which highlight the discriminatory learning of the embedding space obtained from the pre-training paradigm.



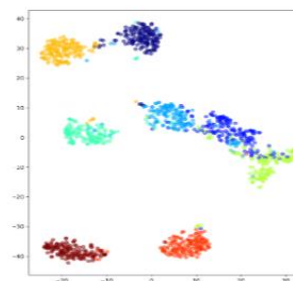
(a) After 40 epochs



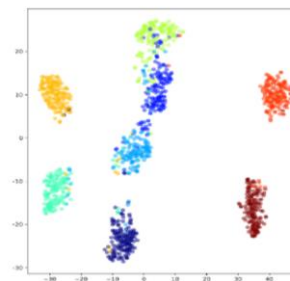
(b) After 80 epochs



(c) After 120 epochs

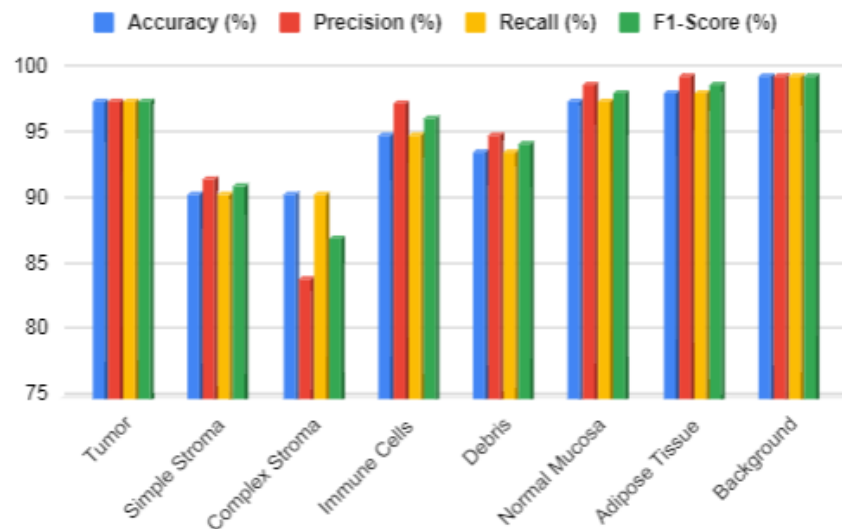


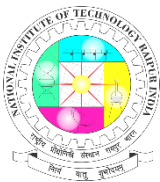
(d) After 160 epochs



(e) After 200 epochs

- The following figure illustrates the class-wise evaluation metrics obtained from the proposed method on the CRCH dataset.

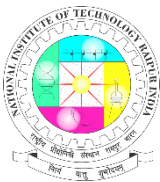




Results and Discussions

- From the following table, which shows the comparison between our method and other state-of-the-art methods on CRCH dataset, it can be seen that our method outperforms the other ones by a significant margin.

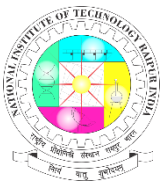
Method	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
Ciampi et al. [3]	79.66	-	-	-
Kather et al. [1]	87.40	-	-	-
Ohata et al. [5]	92.80	-	-	92.12
Wang et al. [2]	92.60	-	92.80	-
Sabol et al. [4]	92.74	92.50	92.76	92.64
Ghosh et al. [6]	92.83	92.83	93.11	92.97
Proposed method	95.22	95.34	95.22	95.26



Results and Discussions

- We have performed an ablation study to quantitatively determine the contribution of the two components of our framework, i.e. (1) a supervised metric learning model and (2) a self-supervised image reconstruction model. The results are shown in the table provided below:

Method	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
Triplet	94.58	94.65	94.58	94.61
Reconstruction	93.39	93.50	93.39	93.41
Triplet+Reconstruction (Proposed)	95.22	95.34	95.22	95.26

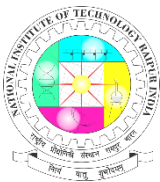


Conclusions

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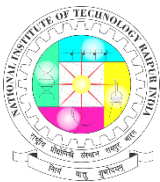


- In this work, we have aimed to propose a novel training strategy for robust representation learning of histopathological images.
- The proposed framework leverages supervised metric learning, for the optimization of embedding space, and self-supervised image reconstruction module, for enforcing pixel-level information learning.
- Thorough analysis of our framework, both qualitative and comparison with existing state-of-the-art works in literature along with suitable ablation study, prove its efficiency for image classification on CRCH dataset.
- Our future work shall be focused on implementation of alternative metric learning losses along with VAE/GAN based models for reconstruction.



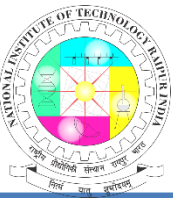
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